Qualcomm Technologies, Inc.

SW6100 External TDM Interface

User Guide

80-74841-55 Rev. AA

December 19, 2025

Revision history

Revision	Date	Description
AA	December 2025	Initial release

Contents

Figures

1 Introduction

1.1 Purpose

This document describes how to configure external TDM interfaces on the SW6100 chipset.

1.2 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `cp armcc armcpp`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, **`copy a:*.* b:`**.

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

1.3 Technical assistance

For assistance or clarification on information in this document, open a technical support case at https://support.qualcomm.com/.

You will need to register for a Qualcomm ID account and your company must have support enabled to access our Case system.

Other systems and support resources are listed on https://qualcomm.com/support.

If you need further assistance, you can send an email to qualcomm.support@qti.qualcomm.com.

2 Audio interface overview and multiplexing

The TDM, MI2S, and AUXPCM interfaces are multiplexed and share GPIOs. Only one can be enabled in each interface simultaneously. SW6100 has three interfaces :

- LPI
  - LPAIF_RXTX Primary – Quaternary
  - LPAIF_VA - Primary – Quinary
  - LPAIF_WSA - Primary – Senary

The following table shows the MI2S/TDM/PCM GPIO assignments.

ADSP interface type	ADSP interface index	HLOS interface name	ADSP audio HW Clock ID	Hardware interface name	On-chip/LPI	GPIO function	CHIP TLMM GPIO # – SCK/SYNC/SD0/SD1	LPI LPASS GPIO # – SCK/SYNC/ SD0/SD1	Wakeup-capable GPIOs
LPAIF_RXTX	Primary	Quaternary	0x206	LPI_I2S0	LPI	2 – lpi_i2s0_sck 2 – lpi_i2s0_ws 2 – lpi_i2s0_data0 2 – lpi_i2s0_data1 2 – lpi_i2s0_data2 3 – lpi_i2s0_data3	GPIO 165/166/167/168/ 169/170	LPASS 0/1/2/3/4/5	GPIO 166/169
LPAIF_VA	Primary	Quinary	0x208	LPI_I2S1	LPI	2 – lpi_i2s1_clk 2 – lpi_i2s1_ws 2 – lpi_i2s1_data0 2 – lpi_i2s1_data1	GPIO 171/172/173/174	LPASS 6/7/8/9	GPIO 171/172/174
LPAIF_WSA	Primary	Senary	0x20B	LPI_I2S2	LPI	1 – lpi_i2s2_clk 1 – lpi_i2s2_ws 1 – lpi_i2s2_data0 1 – lpi_i2s2_data1	GPIO 175/176/180/181	LPASS 10/11/15/16	GPIO 176/181

3 TDM interface

3.1 Overview

TDM is a popular serial bus for interfacing to audio chips. It is a simple data interface without any form of address or device selection. TDM allows multiple channels of audio data to be transmitted over a single data line (see Figure 3-1).

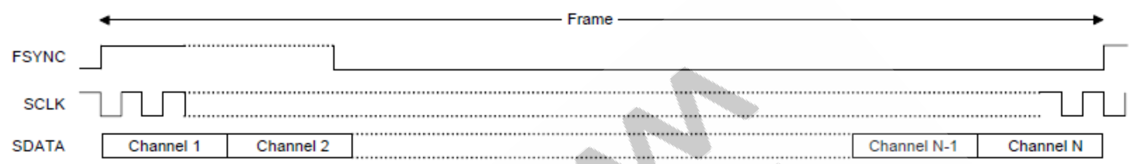


Figure 3-1 TDM interface

TDM involves only one bus master, one transmitter, and one receiver. Either the transmitter or receiver can be a bus master. The TDM interface is comprised of two control clocks, a frame synchronization pulse (FSYNC), a serial clock/bit clock (SCLK), and a serial data line out and line in (see Figure 3-2 and Figure 3-3).

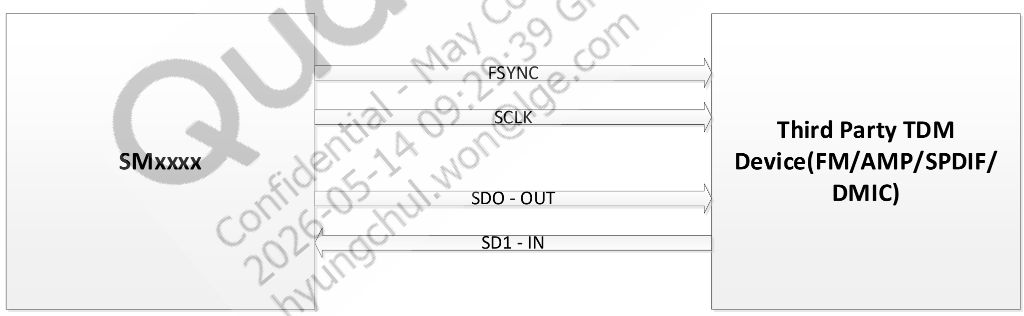


Figure 3-2 SW6100 in Master mode

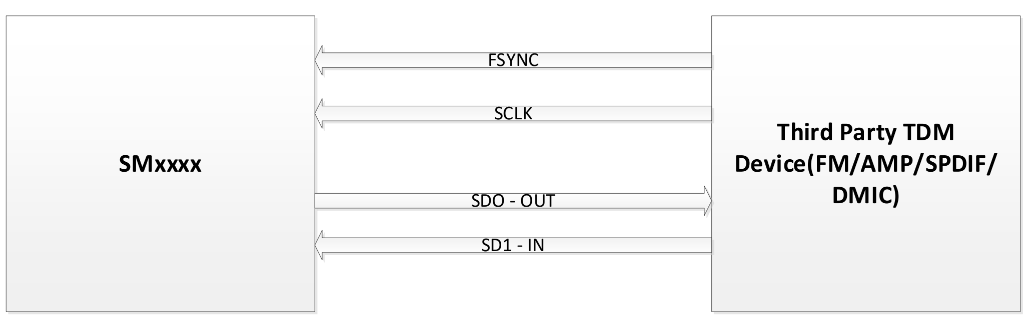


Figure 3-3 SW6100 in Slave mode

3.2 TDM slot and bitwidth configurations

The following figures show the TDM16 and TDM8 configurations with slot widths of 32 and 16 bits. The start byte offset of each slot is shown, which is used to explain the TDM driver code.

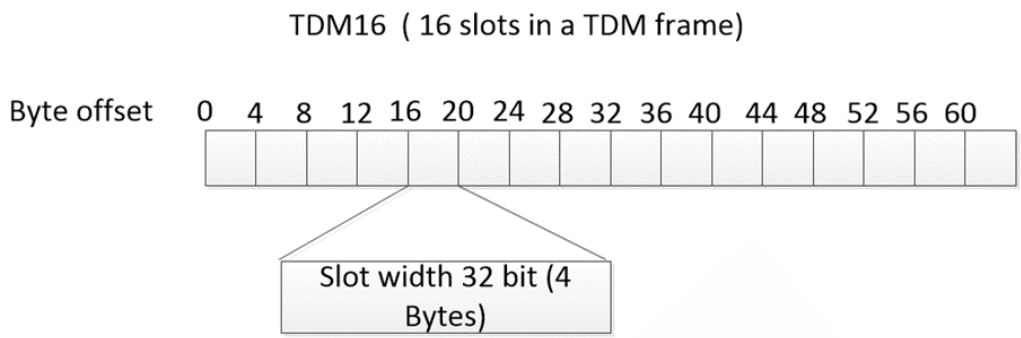


Figure 3-4 TDM16 configuration – 32 bits

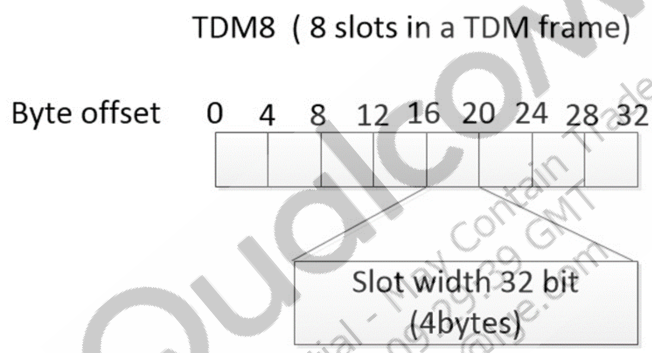


Figure 3-5 TDM8 configuration – 32 bits

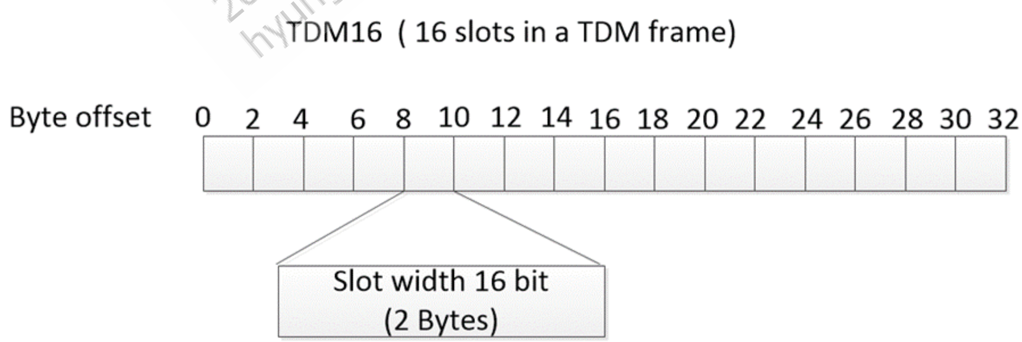


Figure 3-6 TDM16 configuration – 16 bits

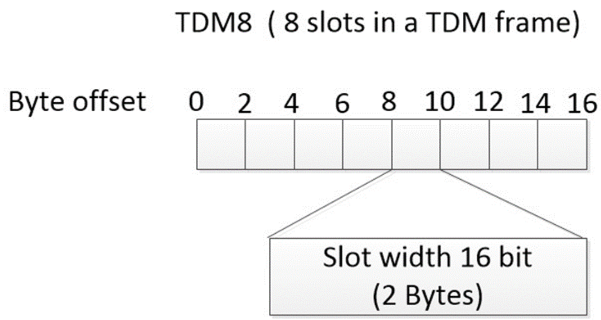


Figure 3-7 TDM8 configuration – 16 bits

3.3 SW6100 features

- Supports three TDM hardware interfaces
- Supports both Master and Slave mode configurations
- One physical Tx serial DOUT lane and one physical Rx serial DIN lane (independent of each other)
- Short sync (sync pulse is a single cycle), long sync (sync pulse is half a frame), and one slot sync (sync pulse is high for the duration of the first slot)
- Frame sync can be inverted (useful for long sync)
- 0 to 2 cycle delays between frame sync pulse edge
- 1 to 512 bits per frame
- 1 to 32 bits per slot (width of serial data in and out can be different)
- 1 to 32 slots per frame, depending on the programmed bit rate and number of bits per slot
- Data sent per slot can be smaller or equal to slot size
  - Tx – If data size < slot size, the remaining bits are padded with zeros
  - Rx – If data size < slot size, all received data after data size is ignored
- Any one of the transit or receive slots are available as the active slot
  - Maximum of all 32 active slots could be used per frame
- Maximum clock frequency supported is 24.576 MHz
- I/O voltage 1.2/1.8 V and drive 2 mA to 16 mA in 2 mA steps

4 TDM software configurations

NOTE: These changes are not part of QTI release builds. The GPIO configuration node has been defined in vienna-audio-overlay.dtsi and the customer needs to enable it in the target.dtsi file based on their configuration.

4.1 GPIO configuration

The machine driver enables/disables the TDM GPIOs whenever the playback or capture use case is started over the TDM interface. The GPIO configuration is specified in the device tree file of the platform and its pinctrl file.

- LPI TDM interface (tertiary, quaternary, senary, quinary and septenary) pinctrl entries are added in the vienna-lpi.dtsi file.

4.1.1 LPI TDM interface GPIO configuration

LPI TDM GPIO active/sleep configurations should be added in the customer target .dtsi file. The following is an example configuration for quaternary TDM interface. Similar configurations should be added for senary and quinary TDM interfaces if the customer is planning to use these interfaces. Ensure that no other interface is using these TDM interface GPIOs for another purpose, and remove all other entries from the device tree.

```
&spf_core_platform {
        cdc_quin_mi2s_gpios: quin_mi2s_pinctrl {
             status = "disabled";
             compatible = "qcom,msm-cdc-pinctrl";
             pinctrl-names = "aud_active", "aud_sleep";
             pinctrl-0 = <&lpi_tdm1_sck_active &lpi_tdm1_ws_active
                        &lpi_tdm1_sd0_active &lpi_i2s1_sd1_active>;
             pinctrl-1 = <&lpi_tdm1_sck_sleep &lpi_tdm1_sd1_active
                        &lpi_tdm1_sd0_sleep &lpi_tdm1_sd1_sleep>;
             qcom,lpi-gpios;
             qcom,tlmm-pins = <171 172 174>;
             #gpio-cells = <0>;
        };
};

&cdc_quin_mi2s_gpios {
    status = "ok";
};
```

```
&vienna_snd {
      qcom,tdm-audio-intf = <1>;
       qcom,quin-mi2s-gpios = <&cdc_quin_mi2s_gpios>;
        qcom,msm-mi2s-master = <1>, <1>, <1>, <1>, <1>, <1>, <1>;
        qcom,mi2s-tdm-is-hw-vote-needed = <1>, <1>, <1>, <1>, <1>, <1>, <1>;
        qcom,tdm-max-slots = <8>;
        qcom,tdm-clk-attribute = <0x1>, <0x1>, <0x1>, <0x1>, <0x1>, <0x1>,
<0x1>;
      qcom,mi2s-clk-attribute = <0x1>, <0x1>, <0x1>, <0x1>, <0x1>, <0x1>,
<0x1>;
};
```

NOTE: The configurations have been defined in vienna-audio-overlay.dtsi and need to be added/enabled in the customer target.dtsi file for the required interface.

Two pinctrl groups are defined as aud_active and aud_sleep for each TDM interface. The msm_common driver enables/disables these pinctrl groups while starting/stopping that particular DAI. The machine driver sets the pins to aud_active when the interface is active and to aud_sleep when not in use.

Some of the LPI GPIOs are wakeup-capable and should be disabled if being used for TDM.

To disable wakeup capability, add the following entry:

```
qcom,tlmm-pins = <TLMM GPIO Number>
```

See Chapter 2 for the wakeup-capable GPIO list.

The pinctrl nodes for quaternary, senary and quinary are added in the vienna-lpi.dtsi file; no changes are needed. The following are example quaternary TDM GPIO configurations:

```
lpi_i2s0_sck_active: lpi_i2s0_sck_active {
        mux {
                pins = "gpio0";
                function = "func1";
        };

        config {
                pins = "gpio0";
                drive-strength = <8>;   /* 8 mA */
                bias-disable;   /* NO PULL */
                output-high;
        };
};

lpi_i2s0_sck_sleep: lpi_i2s0_sck_sleep {
        mux {
                pins = "gpio0";
                function = "func1";
        };

        config {
                pins = "gpio0";
                drive-strength = <2>;   /* 2 mA */
                bias-pull-down;     /* PULL DOWN */
                input-enable;
        };
};

lpi_i2s0_ws_active: lpi_i2s0_ws_active {
        mux {
                pins = "gpio1";
                function = "func1";
        };

        config {
                pins = "gpio1";
                drive-strength = <8>;   /* 8 mA */
                output-high;
        };
};
lpi_i2s0_ws_sleep: lpi_i2s0_ws_sleep {
        mux {
                pins = "gpio1";
                function = "func1";
        };
```

```
        config {
                pins = "gpio1";
                drive-strength = <2>;   /* 2 mA */
                bias-pull-down;       /* PULL DOWN */
                input-enable;
        };
    };

    lpi_i2s0_data0_active: lpi_i2s0_data0_active {
        mux {
                pins = "gpio2";
                function = "func1";
        };

        config {
                pins = "gpio2";
                drive-strength = <8>;   /* 8 mA */
                bias-disable;         /* NO PULL */
        };
    };

    lpi_i2s0_data0_sleep: lpi_i2s0_data0_sleep {
        mux {
                pins = "gpio2";
                function = "func1";
        };

        config {
                pins = "gpio2";
                drive-strength = <2>;   /* 2 mA */
                bias-pull-down;       /* PULL DOWN */
                input-enable;
        };
    };

    lpi_i2s0_data1_active: lpi_i2s0_data1_active {
        mux {
                pins = "gpio3";
                function = "func1";
        };

        config {
                pins = "gpio3";
                drive-strength = <8>;   /* 8 mA */
                bias-disable;         /* NO PULL */
        };
    };
```

NOTE: Data out gpio should be configured as bias-pull-down as shown above where the tdm data line is used as data out.

Each pinctrl node is defined as follows:

- Pins – Specifies the GPIO number.
- Function – Selects the GPIO pad's functional hardware interface. The TDM interface function should be set to MI2S. The pinctrl driver abstracts the hardware, so qua_mi2s can be set to correspond to the TDM interface being used.
- Driver strength – Controls the GPIO pad strength. It is configurable from 2 mA to 16 mA, with a default configuration of 8 mA.
- GPIO pull – Can be configured with internal pull-up, pull-down, keeper function, or pull none (disable all pulls). The required configuration for TDM is bias_disable (NO PULL).

Refer to kernel_platform/common/Documentation/devicetree/bindings/pinctrl/pinctrl-bindings.txt for more information about pinctrl properties.

4.2 TDM Master mode

In SW6100 TDM Master mode, the bit clock frequency should be configured based on:

- Sample rate
- Number of Slots (number of slots ≥ number of channels)
- Slot width (slot width ≥ bit width)

The bit clock frequency is equal to:

```
Sample rate * Number of slots * Slot width
```

Both bit clock and word select (WS) are provided by the MSM™. The WS clock is sourced from the bit clock and set equal to the sample rate. Clock configuration is not needed for WS.

The number of slots can be configured from 2, 4, 6, 8, 10, 12, 14 to 16 based on requirements. Slot width is also fixed in the machine driver and currently set to 4 bytes. However, slot width can be configured from 2, 3 to 4 bytes.

TDM interface clock control, number of channels, number of serial data lines, data line direction bitwidth, and sample rate are handled in LPASS. The apps processor reads the TDM interface clock configuration from ACDB and pushes it to LPASS while starting the use case over the TDM interface. In the case of MSM Master mode, in ACDB, the clock source for the bit clock must be set as internal.

4.3 TDM Slave mode

In Slave mode, both the bit clock and WS are provided by a third-party TDM device. TDM interface clock control, number of channels, number of serial data lines, data line direction bitwidth, and sample rate are handled in LPASS. The apps processor reads the TDM interface configuration from ACDB and pushes it to LPASS while starting the use case over TDM interface. In the case of MSM Slave, in ACDB, the clock source for the bit clock must be set as external.

5 Enabling TDM

By default, release builds have complete configurations for TDM hardware interfaces. To enable the TDM audio hardware interface, make the following changes to .dtsi files:

- vienna-lpi.dtsi – This file contains GPIO configurations from LPI TDM interfaces.
  - Verify that the GPIOs are mapped to the correct TDM hardware interface.
  - Remove GPIO configuration settings if another hardware module is using the same GPIOs as TDM.
- OEM target device tree file – By default, TDM interface GPIO Active/Sleep states are not added to the vienna_snd node. OEMs should update their OEM target .dtsi file with vienna_snd node, adding required TDM interface GPIO Active/Sleep state nodes.

Interface name	GPIO node
Quaternary TDM Interface	qcom,quat-mi2s-gpios
Quinary TDM Interface	qcom,quin-mi2s-gpios
Senary TDM Interface	qcom,sen-mi2s-gpios

The following example shows how to add a septenary TDM interface to the device tree file.

```
&spf_core_platform {
  cdc_quin_mi2s_gpios: quin_mi2s_pinctrl {
      status = "disabled";
      compatible = "qcom,msm-cdc-pinctrl";
      pinctrl-names = "aud_active", "aud_sleep";
      pinctrl-0 = <&lpi_i2s1_sck_active &lpi_i2s1_ws_active
                 &lpi_i2s1_data0_active &lpi_i2s1_data1_active>;
      pinctrl-1 = <&lpi_i2s1_sck_sleep &lpi_i2s1_ws_sleep
                 &lpi_i2s1_data0_sleep &lpi_i2s1_data1_sleep>;
      #gpio-cells = <0>;
  };
};

&cdc_quin_mi2s_gpios {
  status = "ok";
};

 &vienna_snd {
         qcom,tdm-audio-intf = <1>;
```

```
            qcom,quin-mi2s-gpios = <&cdc_quin_mi2s_gpios>;
    };
```

- msm_common.c – This file contains implementation related to parsing the MI2S interface Active/Sleep GPIO configurations. During bootup, ASOC machine probe, the TDM interface GPIO Active/Sleep states will be initialized by going through each node of TDM interface.
- vienna.c – This file contains DAI link configurations for TDM interfaces.
  - By default, all seven TDM DAI links are added for both playback and capture. These are added to the sound card only if the following entry is added to the OEM target device tree file:

    ```
    &vienna_snd {
      qcom,tdm-audio-intf = <1>;
    }
    ```

    If TDM interface is not required, then set above value to 0.
  - OEMs can update these DAI links based on OEM hardware configuration and remove unused DAI links.
  - For fractional sample rate such as 44.1 kHz, audio HW vote should be enabled for Tertiary, Quaternary, Quinary, Senary, and Septenary interfaces so that the ibit clock is sourced from CXO instead of RO. Enable the HW vote field in the OEM target device tree file for the interface.

    ```
    qcom,mi2s-tdm-is-hw-vote-needed = <1>, <0>, <1>, <0>, <0>, <0>, <0>;
    ```

    For example, to enable HW vote for the Quinary interface in the device tree:

    ```
    qcom,mi2s-tdm-is-hw-vote-needed = <1>, <0>, <1>, <0>, <1>, <0>, <0>;
    ```

- msm-audio-defs.h – This header file contains BE DAI link names for TDM interfaces. These names are used by AGM/HAL and ACDB. By default, all TDM interfaces BE DAI link names are added. No changes need to be made to this file.

5.1 Add TDM device to resourcemanager.xml file

For Qualcomm reference WTP/WRD/IDPS platforms, update the corresponding resourcemanager* .xml file by replacing the existing WSA codec device backend names.

For WTP/WRD/IDPS:

vendor/qcom/opensource/pal/configs/vienna/resourcemanager_vienna.xml

The following sections describe example file changes made to vendor/qcom/opensource/pal/configs/vienna/resourcemanager_*.xml on the build system or vendor/etc/audio/sku_vienna/resourcemanager_*.xml on the device for specific use cases.

All TDM interfaces backend names are defined in the vendor/qcom/opensource/audio-kernel/asoc/msm-audio-defs.h file. While updating the resourcemanager_*.xml file for a particular TDM interface, make sure the backend name is matched with the TDM interface backend name defined in the msm-audio-defs.h.

5.1.1 Example 1 – Replacing WSA backend name with TDM backend name for speaker device

This example shows how to replace the WSA speaker AMP backend name with TDM backend names in the resourcemanager.xml file. In this example, the septenary TDM Rx device is added as the speaker device. Other TDM backend devices can be added in a similar manner.

```
<out-device>
             <id>PAL_DEVICE_OUT_SPEAKER</id>
-            <back_end_name>CODEC_DMA-LPAIF_WSA-RX-0</back_end_name>
-            <max_channels>2</max_channels>
-            <channels>2</channels>
-            <samplerate>48000</samplerate>
             <bit_width>16</bit_width>
+           <back_end_name>MI2S-LPAIF_VA-RX-PRIMARY</back_end_name>
+            <max_channels>2</max_channels>
+            <channels>2</channels>
+            <samplerate>48000</samplerate>
           <snd_device_name>speaker</snd_device_name>
               <speaker_protection_enabled>0</speaker_protection_enabled>
```

- max_channels – Maximum number of channels supported over TDM
- channels – Default number of channels
- bit_width – Default bit width
- speaker_protection_enabled
  - 0 – Speaker protection disabled
  - 1 – Speaker protection enabled

5.1.2 Example 2 – Replacing WSA backend name with TDM backend name for handset device

This example shows how to replace the WSA speaker AMP backend name with TDM backend names in the resourcemanager.xml file. In this example, the septenary TDM Rx device is added as the handset device. Other TDM devices can be added in a similar manner.

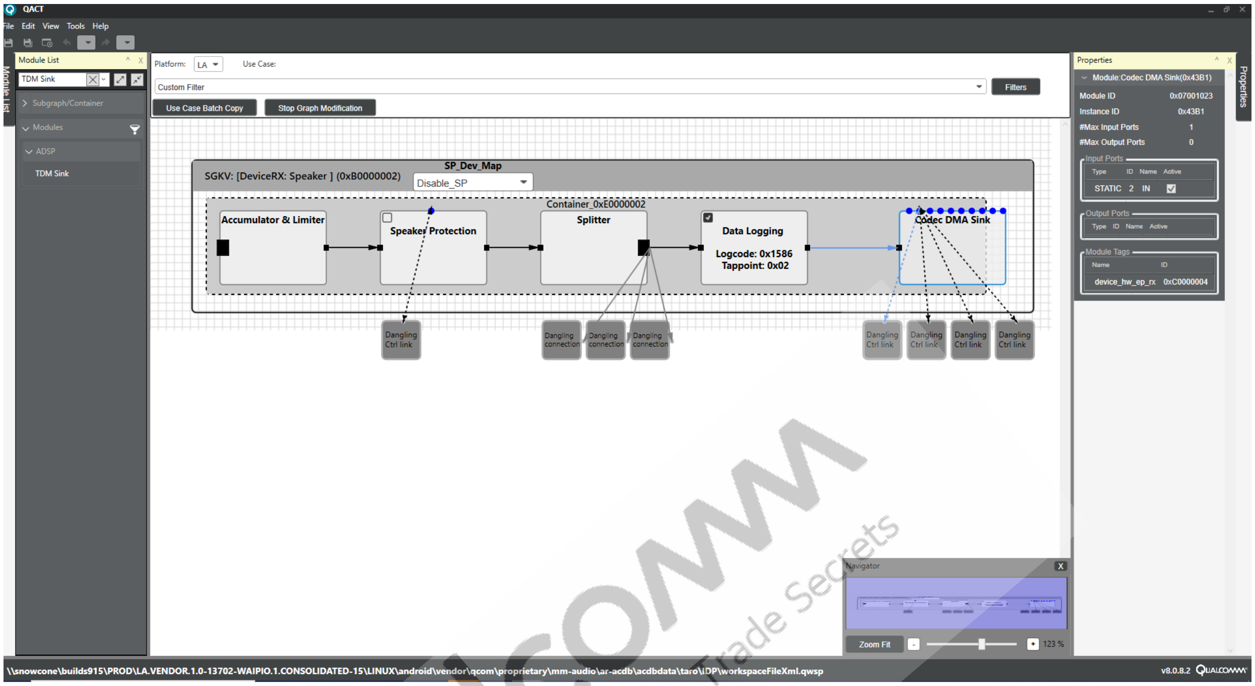
```
    <out-device>
            <id>PAL_DEVICE_OUT_HANDSET</id>
-            <back_end_name>CODEC_DMA-LPAIF_WSA-RX-0</back_end_name>
-            <max_channels>2</max_channels>
-            <channels>1</channels>
+            <back_end_name>MI2S-LPAIF_VA-RX-PRIMARY</back_end_name>
+            <max_channels>2</max_channels>
+            <channels>2</channels>
        <snd_device_name>handset</snd_device_name>
```

5.2 Update use case graph with TDM source/sink devices

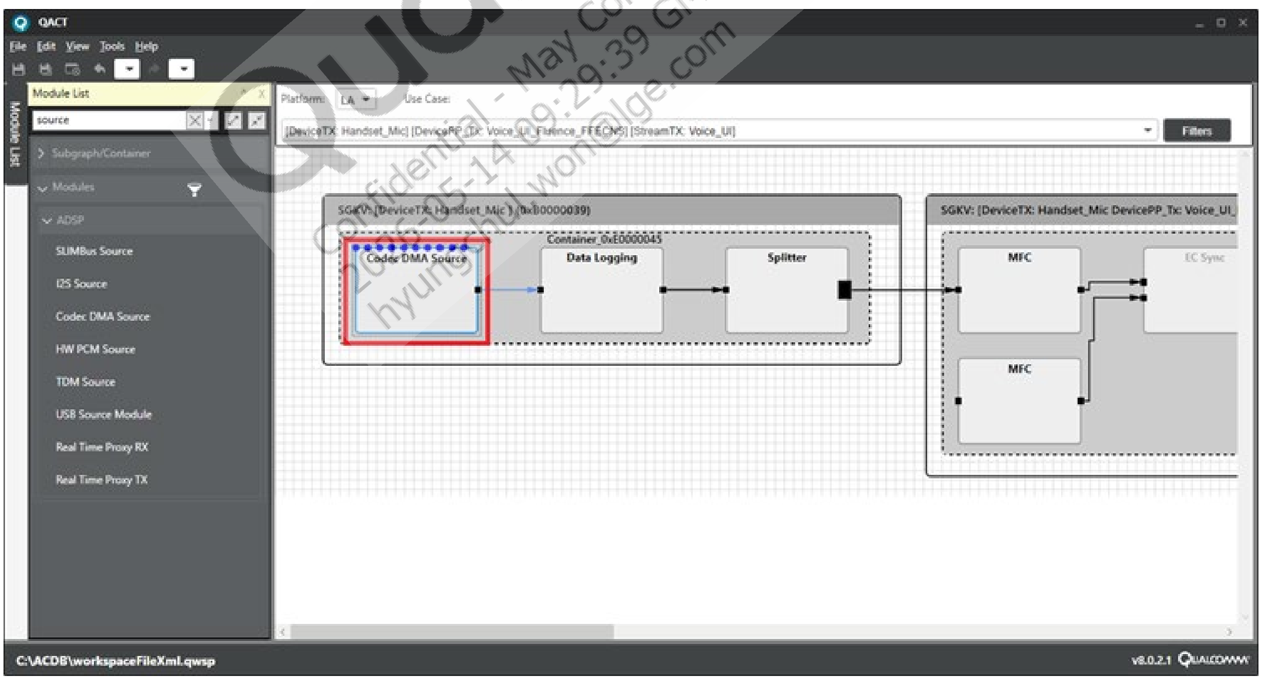
After making the necessary changes in the resourcemanger.xml file, in ACDB, follow these steps to replace the WSA sink/source devices with TDM source/sink devices.

NOTE: Replacing a source or sink module in a device subgraph will affect all use cases where the device is used.

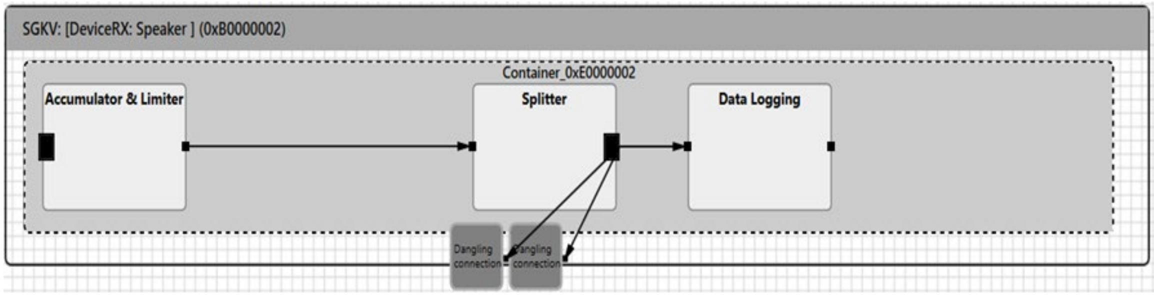
1. In QACT, navigate to a use case containing the desired device subgraph.
2. Select the existing source or sink module.
   - If replacing a sink module, select DeviceRX subgraph.

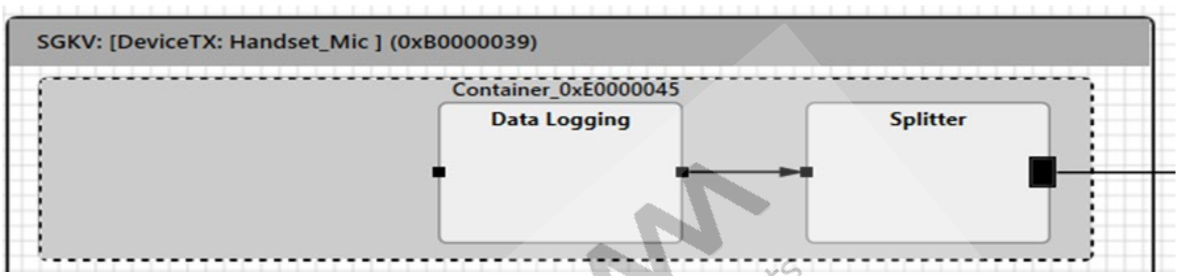


   - If replacing a source module, select DeviceTX subgraph.

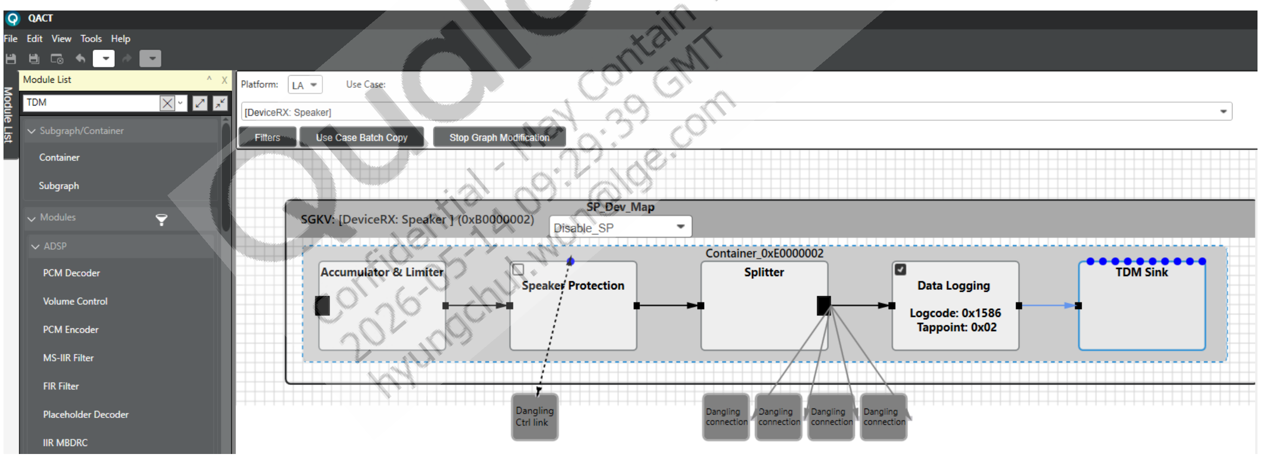


3. Click **Start Graph Modification**, select the sink module to replace, then press **Delete**.



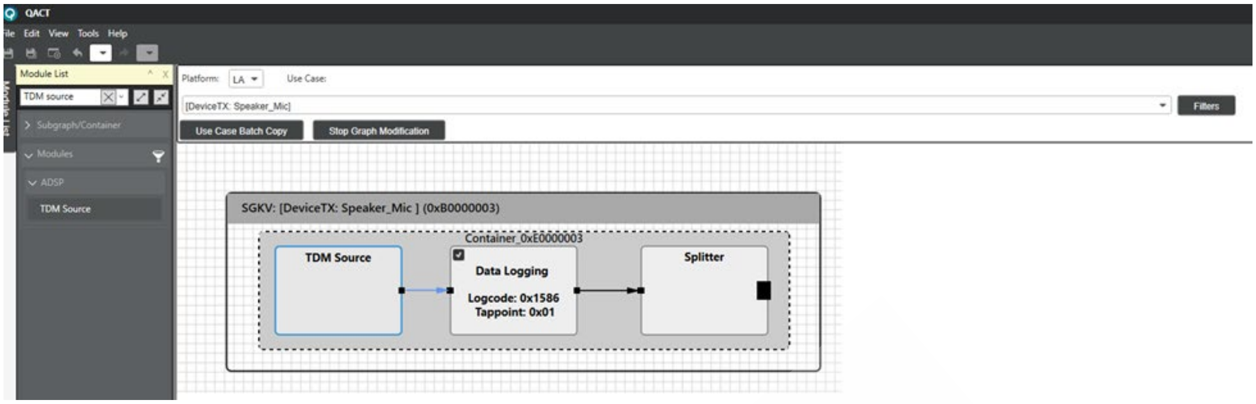


4. From the module list window on the left:
   - Search for sink modules and select TDM Sink module for DeviceRX graph update.



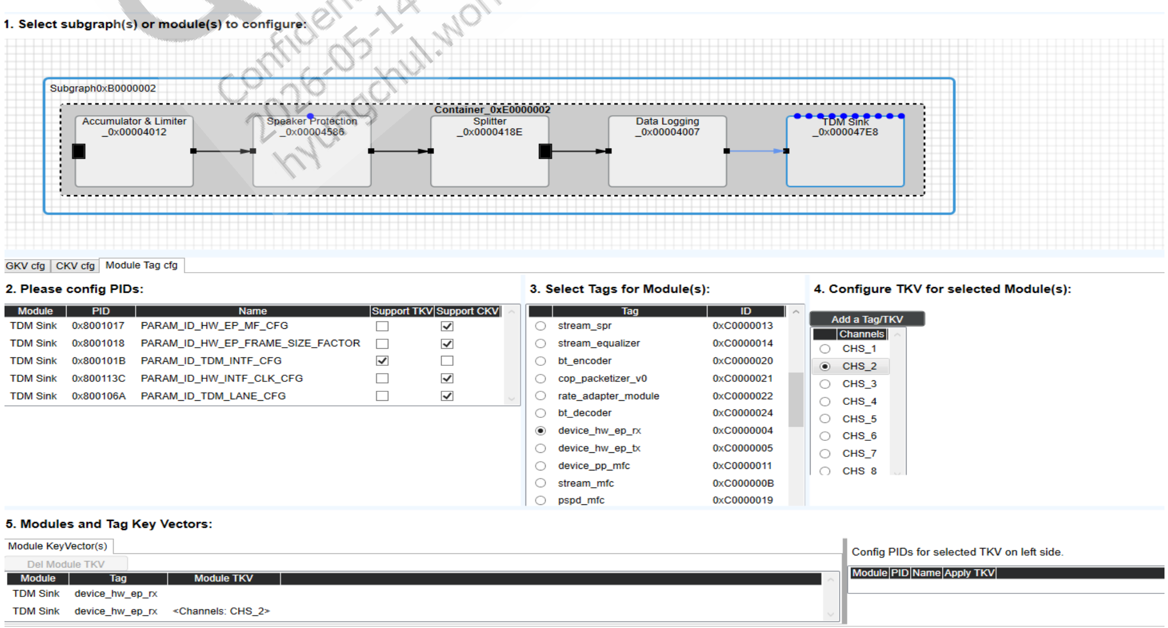
Or

- Search for source modules and select TDM Source module for DeviceTX graph update.



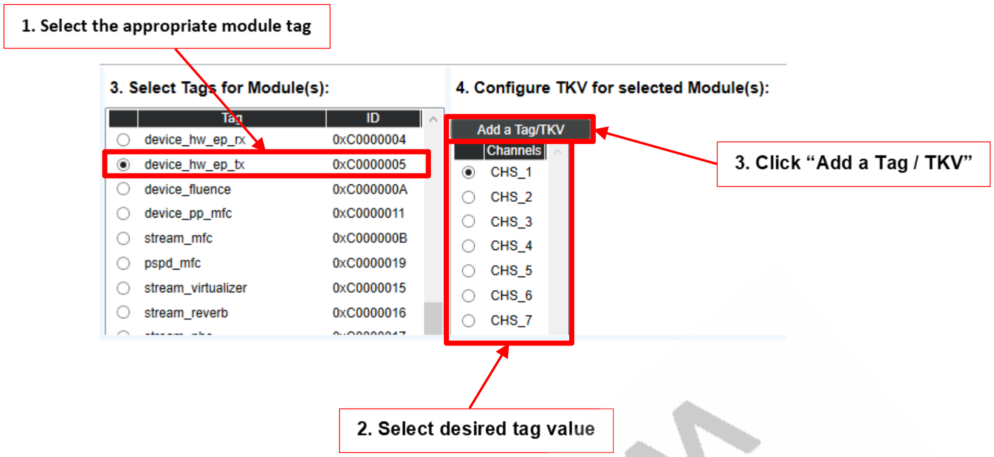
Drag and drop the selected module into the container.

5. Click and drag the new module to connect it to the rest of the subgraph.
6. Before the module parameters can be set, the appropriate module tags need to be assigned. Typically, a module tag is used to control the channel count and corresponding channel mask of a source/sink module. To modify module tags, right-click the subgraph and choose **Configure Keys** to open the Key Configurator. Then click the **Module Tag Cfg** tab.
7. Select the applicable source or sink module. In the Module Tag configurator, change the appropriate PID(s) from "Support CKV" to "Support TKV" to enable TKV control of that group of parameters. The following are supported:
   - Sink – device_hw_ep_rx tag
   - Source – device_hw_ep_tx tag

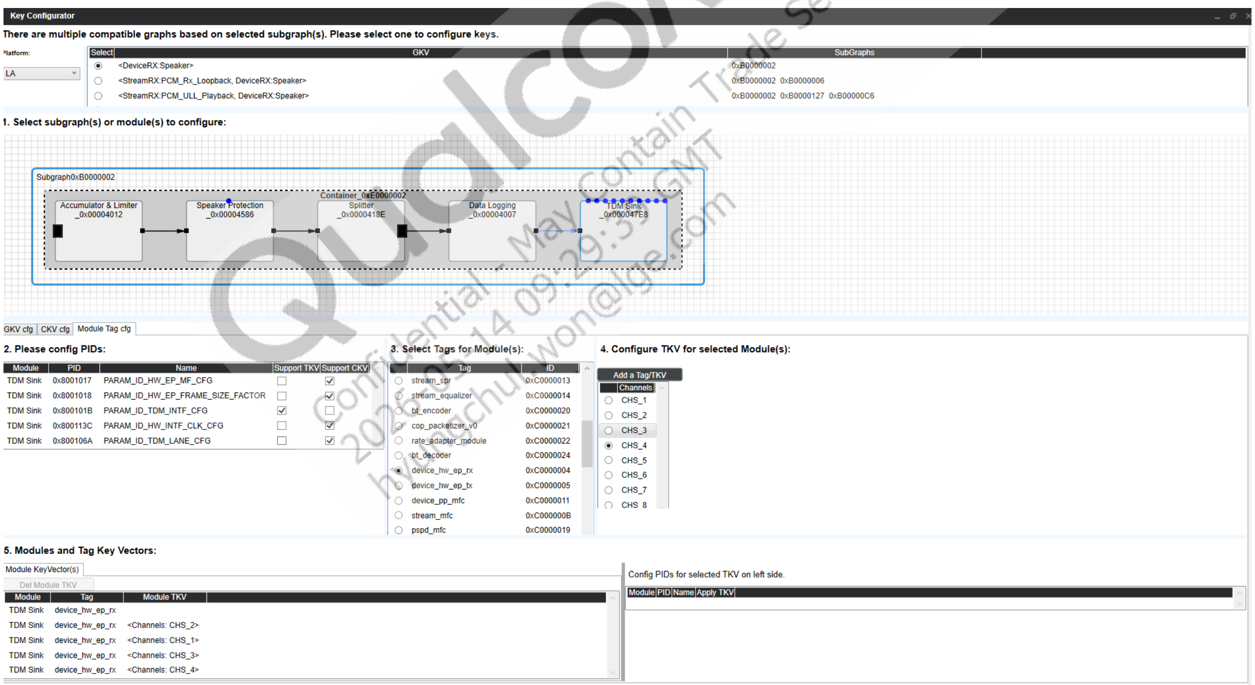


8. Add the appropriate module tags by selecting a tag, then select the TKV, and click **Add a Tag/TKV**. Repeat this step for multiple TKVs, for example, multiple channel counts.

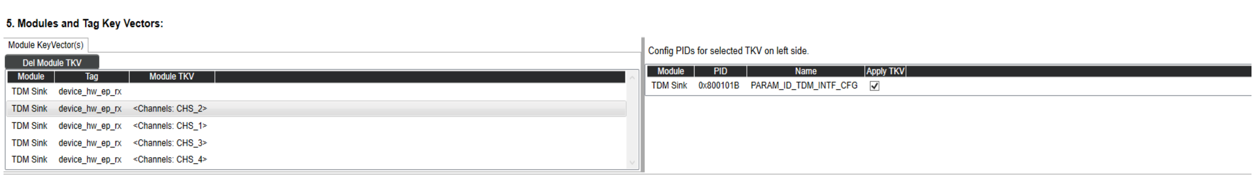
   In the following DeviceTX Graph example, tags are added for channel counts 1-4.



   In the following DeviceRX Graph example, tags are added for channel counts 1-4.

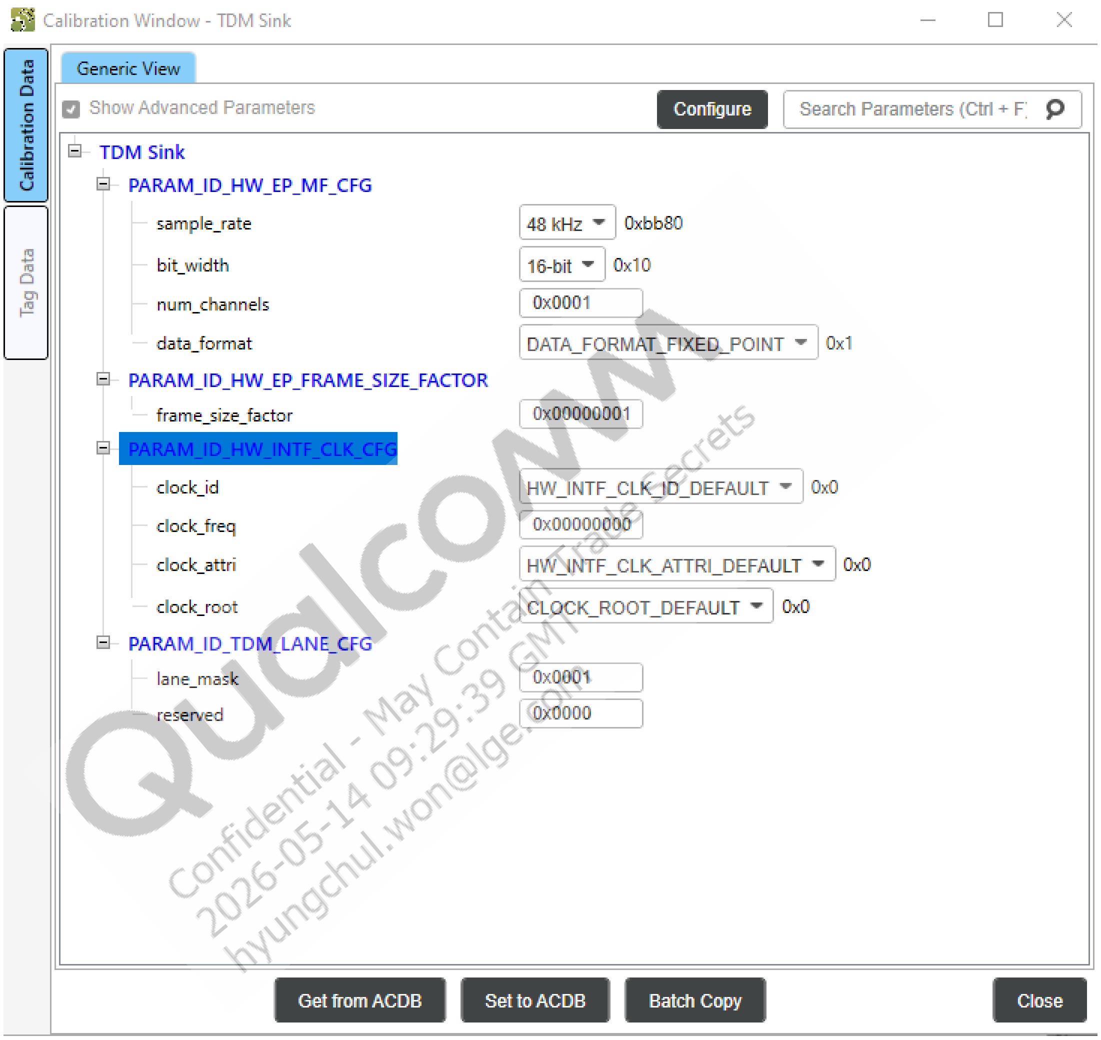


   All TKVs should now be visible in Modules and Tag Key Vectors panel.



TKVs can be selected on the left side, and PIDs configured on the right side for each TKV. Ensure the PID(s) selected in Step 7 is selected for each TKV.

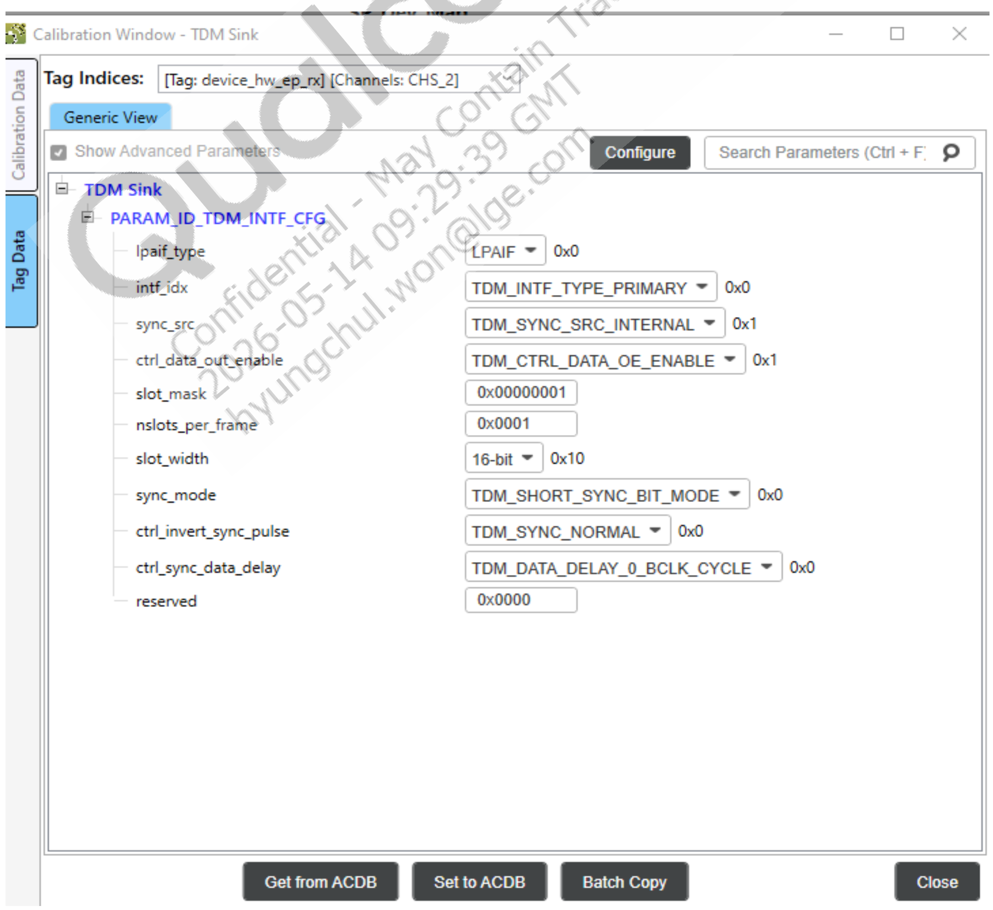
9. Close the Key Configurator.
10. In Graph View, double-click the newly added TDM sink/source module to open the Calibration Window.
11. At the top left, click **Show Advanced Parameters**.



The following hardware endpoint media format configurations from ACDB are not being used; these settings will be overwritten in the Platform Adaptation Layer (PAL) while starting the use case using the resourcemanager.xml:

- PARAM_ID_HW_EP_MF_CFG
  - sample_rate – Sample rate from 8 kHz to 384 kHz
  - bit_width – Bitwidth 16, 24, and 32 bit
  - num_channels – 2/4/6/8
  - data_format – TDM Data format – Fixed Point – Linear PCM
- PARAM_ID_HW_EP_FRAME_SIZE_FACTOR – Use default
- PARAM_ID_HW_INTF_CLK_CFG – Use default
- PARAM_ID_TDM_LANE_CFG – Use default
  - Lane_mask
    - 0x0001 – Selects SD0 for playback if it is TDM sink module
    - 0x0001 – Selects SD0 for capture if it is TDM source module
    - 0x0002 – Selects SD1 for playback if it is TDM sink module
    - 0x0002 – Selects SD1 for capture if it is TDM source module

12. On the left sidebar, click **Tag Data**.



13. Set the module tag of device_hw_ep_tx/device_hw_ep_rx configuration for the different channels.
    - lpaif_type – Select the interface type for TDM interface
      - LPAIF_RXTX – Quaternary TDM interface
      - LPAIF_VA – Quinary TDM interface
      - LPAIF_WSA – Senary interface
    - intf_idx – Select interface index for the TDM interface
      - TDM_INTF_TYPE_PRIMARY – Quaternary TDM interface
      - TDM_INTF_TYPE_PRIMARY – Quinary TDM interface
      - TDM_INTF_TYPE_PRIMARY – Senary TDM interface
    - sync_src – Sync source select
      - Select TDM_SYNC_SRC_INTERNAL for MSM Master mode configuration
      - Select TDM_SYNC_SRC_EXTERNAL for MSM Slave mode configuration
    - ctrl_data_out_enable – TDM block shares data out signal to driver with other masters
      - Select TDM_CTRL_DATA_OE_ENABLE by default
    - slot_mask – Specifies the active slots for channels in 32-bit format
      - 0x00000001 – One active slot/channel at position 0
      - 0x00000002 – One active slot/channel at position 1
      - 0x00000003 – Two active slots/channels at position 0 and 1
      - 0x0000000F – Four active slots/channels at position 0, 1, 2, and 3
    - nslots_per_frame – Specifies number of slots per frame; slot number should be greater than or equal to number of channels
      - 0x0001 – Total number of slots – 1
      - 0x0002 – Total number of slots – 2
      - 0x0008 – Total number of slots – 8
      - 0x0010 – Total number of slots – 16
    - slot_width – Specifies bitwidth of slot; slot bitwidth should be greater than or equal to number of channel bitwidth
      - Supported bitwidth is 16, 24, and 32 bit
    - sync_mode – TDM synchronization mode settings; these settings should be configured based on third-party device hardware requirements
      - Supported modes are TDM_SHORT_SYNC_BIT_MODE, TDM_LONG_SYNC_MODE, TDM_SHORT_SYNC_SLOT_MODE
    - ctrl_invert_sync_pulse – Specifies whether to invert synchronization; these settings should be configured based on third-party device hardware requirements
      - Supported modes are TDM_SYNC _NORMAL, TDM_SYNC_INVERT

- ctrl_sync_data_delay – Specifies number of bit clock cycles for delaying the data for synchronization, supports up to 3-bit clock delays; these settings should be configured based on third-party device hardware requirements
  - Supported settings are TDM_DATA _DELAY_0_BCLK_CYCLE, TDM_DATA _DELAY_2_BCLK_CYCLE, TDM_DATA_DELAY_3_BCLK_CYCLE

6 Logging and debugging

6.1 TDM sink/source playback/capture verification

The AGM test application is used to verify the newly enabled TDM sink or source. This test application is part of the release build. On the device, it is located in vendor/bin folder. To verify MI2S sink/source:

1. Check that the sound card is registered properly with the newly added backend DAI using the following commands:

```
cat /proc/asound/pcm
//Primary TDM
00-30: TDM-LPAIF-RX-PRIMARY msm-stub-rx-30 :  : playback 1
00-31: TDM-LPAIF-TX-PRIMARY msm-stub-tx-31 :  : capture 1
```

   If the backend DAI does not appear in the backend list, there may have been an issue adding either the DAI links or GPIO configurations. Check the kernel logs to make sure there was no error related to DAI links or TDM GPIOs.

2. Update the ACDB playback and graphs with the newly added TDM sink/source device. Also update the backend_conf.xml file with newly added TDM playback/capture backend ID. The backend_conf.xml file is located in the device /vendor/etc folder. The backend name should match the backend DAI name.

```
<device name="TDM-LPAIF-RX-PRIMARY" rate="48000" ch="2" bits="16" />
<device name="TDM-LPAIF-TX-PRIMARY" rate="48000" ch="1" bits="16" />
```

3. Enter the following command to verify playback and capture using adb shell:

```
Playback - agmplay /data/yesterday_48KHz.wav -D 100 -d 100 -i TDM-LPAIF-
RX-PRIMARY
Capture - agmcap /data/yes_cap.wav -D 100 -d 101 -c 2 -r 48000 -b 16 -i
TDM-LPAIF-TX-PRIMARY  -dkv 0xA3000001
```

6.2 Logging

To debug TDM audio playback/recording issues, mixer commands, kernel logs, user space logs, and QXDM Professional™ logs are required.

6.3 Log analysis

This information will be provided in a future revision of this document.

```
// From APSS control using PRM thread.

QDSP6/High
[      audio_hw_clk.c    157]  ADSP: Clk_id :(0x200) Freq : (12288000)
Attr: (0x1) Root: (0x0) Enable: (0x1)

//Check audio hw clock id is correct as per section 2 table

QDSP6/Medium [      audio_hw_clk.c    380]  ADSP: Enable ClkId, incr cnt:
ID: 0x00200 enable count: 1

QDSP6/Medium [      audio_hw_clk.c   1112]  ADSP: SVC:
clk_set_freq_divider_enable: clkid: 0x200 ClkRegId: 0x10109 clk_freq_in_hz:
12288000, divider: 0 clk_enable_cnt: 1

QDSP6/Medium [      audio_hw_clk.c    380]  ADSP: Enable ClkId, incr cnt:
ID: 0x20000 enable count: 1

QDSP6/High [spl_cntr_cmd_handler.c   1296]  ADSP: SC  :10007000: CMD:GPR
cmd: Executing GPR command, opCode(1001006) token(0),  handle_rest 0

QDSP6/High [    pcm_tdm_driver.c    584]  ADSP: PCM TDM Drv: Received hw
media format config: sample rate = 48000, bit width = 16, num channels = 2,
data format = 1

QDSP6/High [    pcm_tdm_driver.c    269]  ADSP: PCM TDM Drv: Received
interface config: intf idx = 0, slots per frame = 8, slot width = 32, sync
src = 1 , slot mask = 3, sync mode = 0, ctrl data out enable= 1, ctrl
invert sync pulse = 0, ctrl sync data delay = 0
```

A References

A.1 Acronyms and terms

Acronym or term	Definition
ASM	Audio stream manager
SoC	System-on-chip
WS	Word select

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